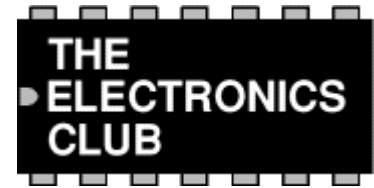


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Variable Resistors

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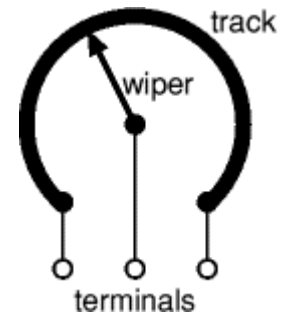


Construction

Variable resistors consist of a resistance **track** with connections at both ends and a **wiper** which moves along the track as you turn the spindle. The track may be made from carbon, cermet (ceramic and metal mixture) or a coil of wire (for low resistances). The track is usually rotary but straight track versions, usually called sliders, are also available.

Variable resistors may be used as a [rheostat](#) with **two** connections (the wiper and just one end of the track) or as a [potentiometer](#) with all **three** connections in use. Miniature versions called [presets](#) are made for setting up circuits which will not require further adjustment.

Variable resistors are often called **potentiometers** in books and catalogues. They are specified by their maximum resistance, linear or logarithmic track, and their physical size. The standard spindle diameter is 6mm.



The resistance and type of track are marked on the body:

4K7 LIN means 4.7 k Ω linear track.

1M LOG means 1 M Ω logarithmic track.

Standard Variable Resistor
Photograph © [Rapid Electronics](#)

Some variable resistors are designed to be mounted directly on the circuit board, but most are for mounting through a hole drilled in the case containing the circuit with stranded wire connecting their terminals to the circuit board.

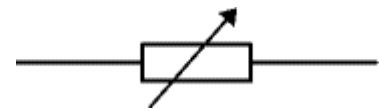
Linear (LIN) and Logarithmic (LOG) tracks

Linear (LIN) track means that the resistance changes at a constant rate as you move the wiper. This is the standard arrangement and you should assume this type is required if a project does not specify the type of track. Presets always have linear tracks.

Logarithmic (LOG) track means that the resistance changes slowly at one end of the track and rapidly at the other end, so halfway along the track is **not** half the total resistance! This arrangement is used for volume (loudness) controls because the human ear has a logarithmic response to loudness so fine control (slow change) is required at low volumes and coarser control (rapid change) at high volumes. It is important to connect the ends of the track the correct way round, if you find that turning the spindle increases the volume rapidly followed by little further change you should swap the connections to the ends of the track.

Rheostat

This is the simplest way of using a variable resistor. **Two terminals** are used: one connected to an end of the track, the other to the moveable wiper. Turning the spindle changes the resistance between the two terminals from zero up to the maximum resistance.



Rheostat Symbol

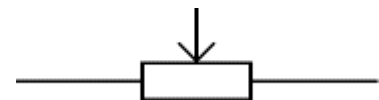
Rheostats are often used to **vary current**, for example to control the brightness of a lamp or the rate at which a capacitor charges.

If the rheostat is mounted on a printed circuit board you may find that all three terminals are connected! However, one of them will be linked to the wiper terminal. This improves the mechanical strength of the mounting but it serves no function electrically.

Potentiometer

Variable resistors used as potentiometers have all **three terminals** connected.

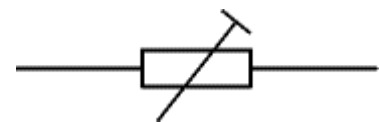
This arrangement is normally used to **vary voltage**, for example to set the switching point of a circuit with a sensor, or control the volume (loudness) in an amplifier circuit. If the terminals at the ends of the track are connected across the power supply then the wiper terminal will provide a voltage which can be varied from zero up to the maximum of the supply.



Potentiometer Symbol

Presets

These are miniature versions of the standard variable resistor. They are designed to be mounted directly onto the circuit board and adjusted only when the circuit is built. For example to set the frequency of an alarm tone or the sensitivity of a light-sensitive circuit. A small screwdriver or similar tool is required to adjust presets.



Preset Symbol

Presets are much cheaper than standard variable resistors so they are sometimes used in projects where a standard variable resistor would normally be used.

Multiturn presets are used where very precise adjustments must be made. The screw must be turned many times (10+) to move the slider from one end of the track to the other, giving very fine control.



Preset
(open style)

Presets
(closed style)

Multiturn preset

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